EFY Note
The source code of this project is available for free download at *source.efymag.com*

Construction and testing

An actual-size PCB layout for the transmitter is shown in Fig. 9 and its components layout in Fig. 10. After assembling the circuit on PCB, connect 5V supply across CON1.

An actual-size PCB layout for receiver is shown in Fig. 11 and its components layout in Fig. 12. After assembling the circuit on PCB, connect another 5V supply across CON3. Use a 5V, 10A contact-rating relay for RL1. As relay RL1 is not PCB-mounted, connect its coil to the PCB where it is marked RL1. Connect the water-pump's power on/off switch across Pole and NO contacts of the relay.

Switch on the transmitter and receiver circuits. Login to *https://io.adafruit.com* and open the dashboard you have already created. Depending on the soil moisture level in the field, you will see the relay status on the dashboard, as shown in Figs 6 and 7. When Relay1 status is 1, motor is on and when Relay1 status is 0, motor is off. The status of the relay will change automatically depending on the soil moisture level in the field.

For testing/checking the signals in transmitter and receiver circuits (Fig. 2 and Fig. 3), the expected signals at different points are listed in Table 1. EFY

PARTS LIST

Semiconductors:	
Board1	- NodeMCU module
SS1	- Soil sensor module
IC1	- HT12E encoder
IC2	- HT12D decoder
D1, D2	- 1N4007 rectifier diode
T1	- BC547 npn transistor
T2, T3	- BC317 npn transistor
LED1-LED4	- 5mm LED
Resistors (all 1/4-watt, ±5% carbon):	
R1	- 1-mega-ohm
R2-R4	- 330-ohm
R5, R7, R9-R11	- 470-ohm
R6	- 47-kilo-ohm
R8	- 4.7-kilo-ohm
Miscellaneous:	
J1	- 3-pin connector with shorting jumper
CON1	- 2-pin connector
CON2	- 4-pin connector
CON3	- 2-pin terminal connector
CON4	- 3-pin connector (not shown in PCB)
CON5	- 3-pin connector
TX1	- 433MHz RF transmitter module
RX1	- 433MHz RF receiver module
RL1	- 5V, 10A single-changeover relay
	- 3V battery/5V DC supply for transmitter
	- 5V DC supply for receiver
ANT.1, ANT.2	- 30cm hook-up wire for antenna


Jitendra Jangir is training officer (Electronics) at National Skill Training Institute (W), Patna, Bihar


V.K. Shukla retired as director at Directorate General of Training, Ministry of Skill Development and Entrepreneurship, government of India